
SoftwareTesting

Welcome! 😊

Who is who and why are we here?

- You are:
 - B.Sc. in Computer Science year 2
 - Any exchange/study abroad (Erasmus) students here?
 - Any one else?
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- I am:
 - Dr. Silvana Togneri MacMahon
 - Previously UAT Test Lead – Retail Banking
 - Medical Device Software Testing Research
 - Room L2.28, 2nd floor, School of Computing Building
 - silvana.macmahon@dcu.ie
 - <http://www.computing.dcu.ie/~smacmahon>
- Why are we here:
 - Module CSC 1029 – Software Testing

The source of all knowledge – Loop!

- Materials and communications will primarily be via Loop, also be email.
- Module Handbook
- Course notes
 - Slides, Recordings, Exercises
 - Recommended reading material
- Assessment - information and results
- Links to related items
- Questions? - How to contact me
 - General Query – Ask a Question on Loop
 - Personal Query - Use Email silvana.macmahon@dcu.ie

Exam and assessment procedure

Continuous Assessment	100%
End of semester examination	0%
Total	100%

- Assessment
 - To be advised and updated on loop – you should regularly consult the information for any updates.
 - The exact format of the assessment will be confirmed later but it is anticipated that all assessments will take the form of a closed book lab exam (quiz) during a timetabled session for the module and will contain a range of multiple choice and text-based questions.
 - Likely breakdown – 3 lab exams: 30%, 40%, 30%
 - Likely timing – Weeks 4, 8 and 12 during the timetabled lab sessions – To be confirmed!
- This module is category 1.
- 1 = A resit is available for all components of the module

Please Note!!!

- Timing of Assessments Labs will be communicated shortly
- There is no lab in Week 1.

Reading List

See Loop for Additional Materials

- It is **strongly** recommended that you obtain a copy of either of the 2 main recommended text books for the duration of the module.
- Andreas Spillner, Tilo Linz, Hans Schaefer, *Software Testing Foundations*, 2nd Edition, RockyNook Books
- Dorothy Graham, Isabel Evans, Erik Van Veenendaal and Rex Black, *Foundations of Software Testing*

Syllabus

Module (**partly**) follows the ISTQB Foundation syllabus

- Fundamentals of testing
 - The test development process
 - What is testing?
 - Categories of test design techniques
 - General testing principles
 - Black-box techniques
 - Fundamental test process
 - White-box techniques
 - The psychology of testing
 - Experience-based techniques
 - Choosing test techniques
- Testing throughout the software life cycle
- Test management
 - Software development models
 - Test organization
 - Test levels
 - Test planning and estimation
 - Test types
 - Test progress monitoring and control
 - Maintenance testing
 - Configuration management
 - Risk and testing
 - Incident management
- Static techniques
 - Static techniques and the test process
- Technology support for testing
 - Review process
 - Types of test technologies
 - Static analysis by tools
 - Effective use of tools: potential benefits and risks
 - Introducing a tool into an organization

ISTQB

International Software Testing Qualifications Board

- ISTQB is the global standard for certification of software testers.
- ISTQB is responsible for the international qualification scheme called "**ISTQB Certified Tester**".
- The qualifications are based on a syllabus, and there is a hierarchy of qualifications and guidelines for accreditation and examination.
- The role of ISTQB is to support a single, universally accepted, international qualification scheme, aimed at software and system testing professionals, by providing the core syllabi and by setting guidelines for accreditation and examination for national boards.
- www.istqb.org
- www.istb.ie

Topics Covered (in no particular order!):

- Fundamentals Of Testing
- Unit Testing Pyunit and Pair Programming
- Test Principles and Psychology of Testing
- Baselines, Exit Criteria & Test Process
- SW Dev Lifecycle Models & Testing
- Test Design Techniques
- Configuration, Risk & Incident Management
- Test Organisation, Planning, Estimation & Control
- Path Coverage
- Testing Software Requirements, Path Coverage
- Low Formality Test Techniques
- Decomposition, Test Levels, Stubs, Drivers
- Data Flow Analysis & Control Flow Graphs
- Debugging

Syllabus

This module will also include additional elements

- Unit Testing
 - What is unit testing?
 - General unit testing principles
 - Unit testing using Junit (Java) / Pyunit (Python)
- Requirements Testing
 - How to test requirements

Help yourself....

Learning in the first instance – Lectures

Pay attention

Don't be passive

Contribute – ask questions, make comments, etc.

Take notes for yourself (and refer to Loop)

Do the exercises set

Read the notes, books, articles

Assessment - Understand concepts and be able to apply them and express your opinions on them

Fundamentals of Testing